ANSWER KEY & MARKING SCHEME SET 1

Chapter 15: **The Making of a Scientist (Footprints Without Feet)** | Class 10 English | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** grows through curiosity, support and effort	Making, not birth.
2	**(c)** taxonomy	Classification.
3	**(a)** show gradual growth	Clear growth.
4	**(d)** it overturned the assumption that they were only ornamental	Challenging assumptions.
5	**(b)** systematic questioning and experiment	Method.
6	**(c)** great discoveries may begin with simple tools	Simple start.
7	**(a)** inspire readers by a real example	Inspirational biography.
8	**(b)** avoiding harm to the animal	Welfare.
9	**(c)** How can schools nurture young scientists like Richard?	Open.
10	**(a)** he tested whether the viceroy truly imitates the monarch	Testing claims.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	He uses familiar examples like butterflies and simple explanations (1) and links ideas to Richard's real experiences (1).	1+1
12	Home gave encouragement and books (1) while the mentor and fairs gave direction and recognition (1).	1+1
13	Patience and perseverance (1) and the willingness to collaborate with mentors (1).	1+1
14	It shows an ordinary child shaped by interest and support (1) so anyone with curiosity and effort can follow the path (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Richard's home supplied books and contacts, and his mother opened doors. Yet his own curiosity and effort did the work. The chapter shows environment and personal drive working together.	1+1+1
16	Collecting and exploring are forms of play. Richard learned through them without pressure. Play can develop skills that later serve research.	1+1+1
17	A hypothesis is a testable guess. Richard's hypothesis was that the viceroy imitates the monarch to avoid predators. He tested it by experiment.	1+1+1
	OR: The writer picks childhood collecting, the book, tagging, and prize-winning projects. Each shows a step upward. By omitting trivial details he creates a clear arc of growth.	1+1+1
18	Richard works openly, ethically and with mentors; Griffin works in secret and misuses his discovery. Richard's science builds knowledge, Griffin's brings isolation. The comparison highlights responsibility.	1+1+1
	OR: It models questioning, testing and evidence-based thinking. Readers see how assumptions are challenged. This builds scientific temper.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Thesis (1). Curiosity: collecting and questions (1). Support: mother and Dr Urquhart (1). Effort: experiments and prizes (1). Conclusion (1).	2+2+1
	OR: Luck: supportive family (1). Effort: experiments (1). Interaction (1). Evidence in the text (1). Verdict (1).	2+2+1
20	Chronological order (1). Simple language (1). Facts and examples (1). Admiring tone (1). Effect on reader (1).	2+2+1
	OR: Introduction (1). What mimicry is (1). Monarch and viceroy (1). Richard's test (1). Closing (1).	2+2+1

Mistakes to avoid:

- Mixing the scientist's name: the chapter is on **Richard H. Ebright**, an American biologist, not on an Indian scientist.
- Saying he became a scientist only because of school; his **mother's support** (trips, telescopes, microscopes) and his collections drove his early curiosity.
- Forgetting the collections: from kindergarten he collected **butterflies**, then rocks, fossils and coins, and became an eager astronomer.
- Missing the turning point: a **book** on butterflies (and a later project on monarch butterflies) shaped his career; the chapter asks how a book changed his life.
- Calling him a famous scientist as a child; he was a shy boy who could not play team sports and so *collected things*.
- Spelling slips: *Ebright, scientist, curiosity, butterfly, telescope*; and tense mixing.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 15: **The Making of a Scientist (Footprints Without Feet)** | Class 10 English | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** It tells events while explaining scientific ideas	Blend.
2	**(a)** science-fair success	An achievement.
3	**(a)** shows few failures or hardships	Selective success story.
4	**(a)** not enough without personal exploration	Complementary.
5	**(c)** assumption versus experimental result	Contrast in the chapter.
6	**(b)** a collaborative enterprise	Collaboration.
7	**(b)** admiring	Admiring.
8	**(c)** to imply is to suggest; to infer is to conclude	Distinction.
9	**(d)** a gradual shaping by experience and encouragement	Gradual.
10	**(b)** not supported since the text stresses effort and curiosity	Not supported.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It can be observed in nature and tested by simple experiments (1) and it links to ideas of survival and evolution (1).	1+1
12	Observation raises questions and theory suggests explanations (1) and experiment links the two (1).	1+1
13	Each project begins with a question such as whether the viceroy mimics the monarch (1) and answering it leads to new knowledge (1).	1+1
14	By encouraging projects based on local observation (1) and by connecting students to experts and fairs (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Dr Urquhart supplied direction and a real project. The mother connected the child to him. Mentors turn private curiosity into public contribution.	1+1+1
16	Indian students can follow their curiosity through nature and school projects. Science fairs and mentors are available in schools and colleges. The chapter encourages them to start early.	1+1+1
17	It gives selected facts and an interpretive view. It may simplify complex research. For full accuracy one must read the original studies.	1+1+1
	OR: Observing butterflies and pupae led to knowledge of mimicry and hormones. No advanced machines were used at first. Careful observation and thought were enough.	1+1+1
18	By showing the long sequence of projects across years. Each success leads to a harder question. The pattern reveals persistence without naming it.	1+1+1
	OR: It would be more personal with emotions and doubts. It might show setbacks more clearly. The third-person voice gives distance and admiration.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Introduction (1). Lessons from Richard (1). Role of teachers and labs (1). Role of projects and fairs (1). Conclusion (1).	2+2+1
	OR: Butterflies in a backyard (1). Hobbies to research (1). Understanding nature (1). Application to health and DNA (1). Conclusion (1).	2+2+1
20	Observation: collecting (1). Experimental: viceroy and pupa (1). Theoretical: DNA studies (1). How they connect (1). Importance for science (1).	2+2+1
	OR: Format (1). Questions about his childhood (1). Her role (1). Advice to parents (1). Closing (1).	2+2+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 15: **The Making of a Scientist (Footprints Without Feet)** | Class 10 English | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** new work builds on earlier work	Builds on others.
2	**(b)** starting projects of his own after reading	Self-directed.
3	**(c)** an active seeker	Active.
4	**(d)** mention of setbacks Richard faced	Balanced.
5	**(a)** that predators learn to avoid the monarch's appearance	Mimicry logic.
6	**(d)** early interest can grow with care	Growth.
7	**(a)** Nurture curiosity and it may grow into discovery	Moral.
8	**(a)** If his mother had not written, Richard might not have met Dr Urquhart.	Past unreal.
9	**(a)** is a testable proposed explanation	Hypothesis.
10	**(d)** The judge said that his work was original.	Backshift.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It lets readers see growth step by step (1) but may oversimplify a complicated career (1).	1+1
12	A belief that spots are mere decoration was tested (1) and shown wrong, proving the need to verify assumptions (1).	1+1
13	His early interest and support helped (1) but results came from sustained effort, so success was earned, not inevitable (1).	1+1
14	Young researchers bring fresh questions (1) and their discoveries can add real data (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Admiration encourages emulation but may hide difficulties. The reader is inspired and sees Richard as a model. A balanced account might also show failures.	1+1+1
16	After Richard read the book, he became curious and began tagging butterflies. The subordinate clause gives the first event.	1+1+1
17	Tagging yields data about migration. Data allows testing of ideas and finding patterns. Richard's careful records contributed to knowledge.	1+1+1
	OR: Tagging and experiments affect living animals. Scientists must avoid harm and respect life. The chapter implies care by describing non-destructive tagging.	1+1+1
18	The teacher said to Richard, 'Your experiment is interesting.' Pronouns and tense change to match the speaker.	1+1+1
	OR: Curiosity makes one ask questions; humility accepts that one may be wrong. Testing assumptions requires humility. Richard's experiments show both.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	State view (1). What it shows (1). What it omits: failures, cost (1). Value for inspiration (1). Balanced conclusion (1).	2+2+1
	OR: Platform (1). Feedback (1). Motivation (1). Possible pressure (1). Conclusion (1).	2+2+1
20	Introduction (1). Habits like observation (1). Projects (1). Seeking mentors (1). Conclusion (1).	2+2+1
	OR: Title and author (1). Summary (1). Strengths (1). Weaknesses (1). Recommendation (1).	2+2+1

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